

Comparing Sentiment on Political Blogs on the Dark Web and the Surface Web

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1 Introduction

The rise of social media has enabled users to broaden their social networks and share opinions more widely than ever before. This development has given rise to so-called "influencers", individuals with large online followings who are able to earn a living through their presence on social media platforms. Where earlier internet users interacted mainly through chats and forums, platforms such as Instagram and TikTok have now become more dominant. Still, traditional forums remain popular, with Reddit standing out as one of the most widely used. Reddit allows users to post anonymously, comment, and vote on content within various topic-based communities. This anonymity, while valuable for open discussion, also creates an environment in which hate speech can thrive. We will define hate speech here as “any kind of communication in speech, writing or behaviour that attacks or uses pejorative or discriminatory language with reference to a person or a group on the basis of who they are—in other words, on their religion, ethnicity, nationality, race, colour, descent, gender or other identity factor” [13].

One of the more notorious anonymous forums is 4chan, an imageboard where users can start discussions by posting images or memes without needing to create an account. The forum’s anonymity and lack of moderation have made it infamous for disturbing and offensive content. The origin of these types of boards lies in Japan, where a student created the original forum known as 2channel (or 2ch). As a backup to this platform, fans later created 2chan. Eventually, a Western version emerged, named 4chan, which became the most popular English-language imageboard. Over the years, several similar boards have appeared and disappeared, such as 8chan, 8kun, 7chan, Endchan, and Dreamchan. These platforms often divide their content across specific boards. For example, 4chan features /b/ for general discussion, /sp/ for sports, and /int/ for international topics. The board of interest for this research is /pol/, short for “politically incorrect,” which has been the subject of prior research by Hine et al. [8].

The combination of anonymous imageboards and the added anonymity provided by tools such as the Tor browser complicates efforts to identify individuals behind the posts. Still, these platforms frequently act as echo chambers that may encourage further radicalization. As Topor notes in his dissertation, racist and antisemitic expressions are generally not accepted in mainstream society, and as a result, such ideologies often shift to the outer edges of the internet. Topor [15] distinguishes between two types of antisemitism: left-wing antisemitism, which is typically characterized by criticism of Israeli policy or anti-Zionism, and right-wing antisemitism, which is grounded in beliefs that Jewish people are inferior to white or Muslim populations.

This research focuses on the latter, right-wing antisemitism, and aims to expand upon the work by Hine et al. [8]. It conducts a similar content analysis of the politically incorrect board (/pol/) on Endchan, an imageboard accessed through the dark web, and compares this to the content found on 4chan.

2 Literature Research

The internet is often divided into three layers: the surface web, the deep web, and the dark web. Kavallieros et al. [1] explain that while the surface web is publicly accessible and indexed by search engines like Google, the deep

web includes content that cannot be accessed through standard search engines but is still available to the general public using specific methods. The dark web, a subset of the deep web, requires specialized browsers such as Tor to access, and is strongly associated with anonymity and illicit activity. Notably, the dark web is not synonymous with the deep web, and approximately half of its content is estimated to be illegal (Chap. 2, [1]). An important feature of .onion sites is their mutual anonymity: the identity of both the visitor and the host remain hidden, unlike mainstream websites which use tracking tools such as cookies, IP addresses, and geolocation to monitor users [5].

As hate speech has become more prevalent online, the amount of research into online hate has also increased. De Gibert et al. [6] note the difficulty of manually labeling hate speech and propose a framework based on sentence-level classification. Analyzing content from the white supremacist forum Stormfront.org, they introduced three annotation categories: HATE (explicit hate speech), noHATE (non-hateful though potentially offensive language), and SKIP (ambiguous content). While the forum is filled with offensive rhetoric, not all such rhetoric qualifies as hate speech according to their criteria. Their classification method provides a useful foundation for distinguishing between offensive and hateful content in ideologically extreme environments.

In response to the growing need for scalable hate speech detection, machine learning tools such as the Perspective API have been developed. This API assigns toxicity scores to comments based on their predicted impact on a conversation. However, in spaces like the one examined in this research, where hate and toxicity are expected to dominate, such tools may not offer sufficient granularity or contextual nuance [14].

Online hate speech has tangible effects beyond digital spaces. Castaño-Pulgarín et al. [4] have shown that online hate can contribute to real-world harm, including violent incidents such as mass shootings [12], as well as the spread of misinformation. Despite the broader public concern around the dark web, much of the academic research has focused on illegal markets, particularly drug sales [11]. Kajan emphasizes that fears around censorship or backlash may push individuals with extreme views to express themselves more freely on the dark web. Using content analysis on a Finnish darknet message board, Kajan found that discussions were frequently initiated using links to real-world news articles and that these often devolved into threads filled with hate speech.

Ylä-Anttila et al. [16] examined the use of political hate speech on the Finnish imageboard Overboard, training a convolutional neural network to detect political messages. Their findings suggest that around ten percent of the content was political in nature, with most of that leaning far-right, even on parts of the board not designated for political discussion. Similarly, Zannettou et al. [17] studied Gab, a self-proclaimed free speech platform that has become known for hosting racist and hateful content. They reported that approximately 5.4 percent of posts contained hate words, as identified using the Hatebase dictionary, a resource also employed by Hine et al. [8].

Hine et al. conducted a more focused investigation into 4chan’s politically incorrect board, /pol/, which they described as a long-standing hub of internet extremism. Their work analyzed both the userbase and the content of discussions on /pol/, and compared these to other 4chan boards such as Sports and International. They found that 12 percent of posts on /pol/ contained hate-related terms, while Sports and International had significantly lower rates of 6.3 and 7.3 percent respectively. Their use of the Hatebase dictionary in combination with the NLTK framework highlighted the limitations of such tools, particularly in contexts where sarcasm, trolling, or coded language may complicate hate speech detection.

This research builds on the above studies by focusing on content from both the surface web and the dark web. In particular, it examines the /pol/ board on Endchan, an imageboard accessible via the Tor network, and draws comparative insights with more mainstream platforms like 4chan. By conducting a semantic analysis of user content across these two platforms, this work aims to contribute to the broader understanding of how hate speech manifests in some of the internet’s most unregulated and anonymous spaces.

3 Method

In this study, the goal was to compare the sentiment expressed in political discussions across two forums: Endchan, a spin-off imageboard from 4chan that is also accessible via the dark web, and 4chan, a similar imageboard hosted on the surface web. Specifically, we focused on the 'Politically Incorrect' boards of both sites, where users engage in discussions on political topics. Our goal was to analyze differences in topics and sentiment between the two platforms and understand whether the anonymity and accessibility of the dark web influence the nature of political discussions.

3.1 Crawler and Scraper

To conduct this research, data from both 'Politically Incorrect' boards was collected using a self-built web crawler and scraper.

The crawler was built to first crawl all pages of the /pol/ board, going from 1 through 14. Because the pages are indexed in the url with '/pol/1', '/pol/2', etc., the crawler could easily scroll through these pages by adjusting the last number. Endchan required access through the Tor network, which presents additional challenges, such as maintaining anonymity and overcoming access restrictions (Biryukov et al. [2]).

Next, the scraper collected the specific, desired data from the board. This included the thread IDs and their content, the subpost IDs and content, as well as the associated thread ID and timestamps. Since the threads were collapsed, the scraper also saved the link to unfold these threads. This was put in a CSV file, an example is shown in Figure 4b. This example shows the threads from Endchan, but the same had been done for 4chan. Due to the cloud verification of 4chan, a new library (*playwright*) was applied to simulate launching a new browser. As mentioned, the scraper also collected the links to unfold the threads. These links were put in a second crawler, who recursively crawled these pages and the posts within these pages to gather all posts within a thread. To be able to differentiate between different threads, the `thread_id` was also saved in this new csv; 'posts.csv', which is shown in Figure 4a.

3.2 Data Preprocessing

The data thus consisted of 4 different CSV files. Two for the 4chan threads and posts, and two for the Endchan threads and posts. The next step was preprocessing the data to remove non-relevant content such as spam and duplicate posts using the NLTK library. This started with tokenizing the text, removing stopwords and punctuation, and lemmatizing linguistic variations to be able to perform a meaningful analysis.

3.3 Analysis

Topic Modeling First, topic modeling was done to compare the different types of topics discussed on the two forums. As mentioned above, the crawler collects threads, and all posts within a thread. In other words, the thread will define the topic of all posts underneath this thread. Therefore, topic modeling was performed on the threads data. Several models were applied, starting with a basic LDA model from the scikit-learn library. For this LDA model, a predefined number of topics was set to 5. The model generated a DataFrame where the first column contains the words for topic 0, and the second column the weights per word. The third and fourth column contains this information for topic 1, fifth and sixth for topic 2, etc. The model does not assign topic, it simply clusters the words together. For comparison, the second model that was used for topic modeling was BERTopic [7]. This model also clusters words into groups for a specific topic, without assigning the topic itself. Opposite to LDA, BERTopic decides on the number of topics on its own. Because neither of the algorithms will assign topics to

these clusters, a third model was applied. DarkBERT is a model specifically trained on data obtained on the Dark Web [10]. It was applied as an unmasking algorithm, to assign topics to the given input of word clusters. Finally, the topic classification model provided by TweetNLP was used to assign topic labels directly to the thread data.

Sentiment Analysis After performing the topic modeling on the threads of both forums, we conducted a sentiment analysis on the posts of both forums. Two different methods for sentiment analysis were applied; VADER, an NLTK model for sentiment analysis specifically designed for analyzing social media texts [9], and TweetNLP, a Python library that provides access to transformer-based language models, including a RoBERTa-based model for sentiment classification [3]. Both libraries categorized content as positive, neutral, or negative. As TweetNLP offers several classification models, such as emotion classification, hate detection, irony detection, and offensive language detection, these models were implemented as well to gain deeper insight into the sources of sentiment. For both datasets, sentiment labels, emotion labels, hate speech labels, irony labels, and offensive language labels were added, along with the corresponding probability scores produced by each model. These probability scores indicate the model’s confidence in the assigned labels. Finally, a Streamlit application was developed to visualize the sentiment, emotion, irony, hate speech, and offensive language labels along with their associated probability scores.

Named Entity Recognition Lastly, Named-Entity-Recognition was performed in order to be able to perform a sentiment analysis on the different entities from the posts. This was done by using spaCy’s pre-trained NER model *‘en-core-web-sm’*. For the NER analysis, the context of each entity was taken into account, which was defined by 50 characters around the entity; so 50 characters in front of the entity, if possible, and 50 characters after the last letter of the entity. The sentiment category was set as ‘negative’ for a sentiment score below 0.05, ‘positive’ for a sentiment score above 0.05, and ‘neutral’ in between these two scores.

Comparison After obtaining the sentiment visualizations for both datasets, the results were compared to identify patterns and differences between Endchan and 4chan. Some key aspects of this comparison included:

- Overall sentiment distribution across the two platforms.
- Variability in sentiment for different entities.

The entities could be combined into topics to evaluate topic sentiment, for instance ‘the United States’, ‘America’, and ‘US’, could be considered of the same topic.

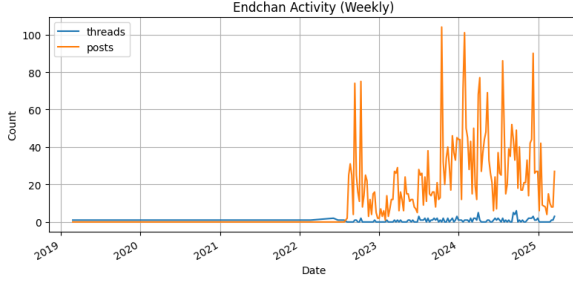
4 Results

4.1 Initial Data Exploration

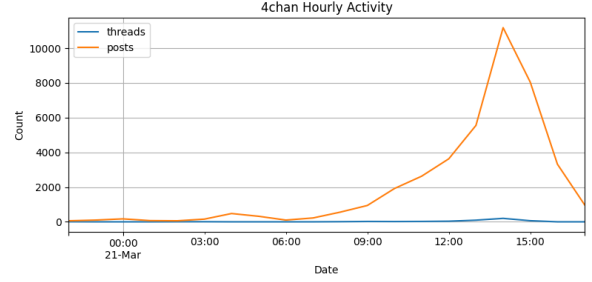
After crawling and scraping the websites, we obtained 10 HTML pages for 4chan, containing 470 threads and 40,459 posts, and 14 HTML pages for Endchan, containing 135 threads and 6,014 posts. Since we also collected the timestamps of the posts from both forums, we were able to visualize the number of posts and threads over time to get an idea of the activity on each forum. From this, we observed that the 4chan data was mainly collected over the span of one day, while the Endchan data covers multiple years (Figure 1). This suggests that Endchan is a smaller and less active forum compared to 4chan. Finally, we noticed a clear increase in activity in the early afternoon, which corresponds to the start of a new day in the US. This is relevant because the imageboards are English-dominant, and we therefore expect a significant amount of US-based activity.

4.2 Topic Modeling

After first applying a basic NLTK modeling, we found that this algorithm was not able to appoint topics to the word clusters. The word clusters were displayed, which meant that a manual human interpretation of the cluster was necessary to assign topics. However, the clusters were filled with random words with no correlation to each



(a) Endchan activity (Weekly)



(b) 4chan activity on the 21st of March (Hourly)

Fig. 1

Topic	4chan				
0	https (128.2)	trump (120.0)	com (101.2)	www (55.4)	year (14.0)
1	us (42.1)	go (35.0)	europe (30.2)	's (26.4)	like (18.5)
2	make (70.4)	like (45.9)	get (23.1)	year (18.4)	would (7.3)
3	even (39.5)	www (36.0)	uk (33.2)	new (26.2)	's (19.8)
4	white (78.5)	would (61.3)	people (36.1)	world (29.7)	literally (27.2)
Topic	Endchan				
0	com (97.2)	https (91.9)	one (7.5)	like (6.7)	many (5.8)
1	jews (125.1)	jewish (98.7)	world (15.0)	would (8.6)	even (7.7)
2	's (76.3)	many (17.5)	like (17.4)	even (4.8)	one (4.3)
3	world (65.0)	's (28.1)	time (22.7)	people (16.3)	one (11.9)
4	people (61.2)	would (41.8)	like (36.3)	one (26.3)	even (22.8)

Table 1: LDA Topics and weights

Forum	Topic Number	Name	Top Words (Representation)
4chan	0	0_white_search_google_anti	white, search, google, anti, nobody, men, family
	1	1_recall_european_would_tesla	recall, european, would, tesla, glue, musk, etc.
	2	2_notice_courts_something_libs	notice, courts, something, libs, slated, accelerated
	3	3_step_us_great_america	step, us, great, america, every, commonwealth
	4	4_faces_sold_want_even	faces, sold, want, even, child, friends, make
Endchan	0	0_media_thread_people_https	media, thread, people, https, like, ai, moe
	1	1_jewish_jews_make_jew	jewish, jews, make, jew, ritual, butchered, history

Table 2: BERTopic Topics and representative words

other, and some words even appeared in multiple clusters (Table 1). A similar result was found for BERTopic (Table 2). Then, DarkBERT was applied to unmask the words and summarize them into one topic. However, if all words of one topic were used as input for DarkBERT, the algorithm would not return any topic as an answer. After some testing, it appeared that the model would only be able to provide an (convincing) answer when the input was 5-6 words. The results of the unmasking tool would often be similar to the word input, not as a topic but specifically one or two of the input words. For instance, if the input words would be 'poiliev', 'russian', 'care', 'peac', 'poll', and 'watch', the algorithm would return the topic 'watch', which is not very accurate for this group of words. Lastly, the topics with -1 are disregarded as these represent the outliers in the data that the model could not classify. For the Endchan data, this was the case for 79 threads, which is over half of the threads in the data.

Forum	Input Words	Top Prediction	Other Notables
4chan	white, search, google, anti, nobody, men	people (0.44)	men, white, women, evil
	recall, european, would, tesla, glue, musk	products (0.05)	drugs, things, " "
	notice, courts, something, libs, slated, accelerationism	things (0.14)	people, shit, names
	step, us, great, america, every, commonwealth	money (0.08)	people, countries, things
	faces, sold, want, even, child, friends	things (0.13)	people, animals, humans
Endchan	media, thread, people, https, like, ai	media (0.14)	content, people, links
	jewish, jews, make, jew, ritual, butchered	jew (0.57)	Jews, Jewish, " "

Table 3: DarkBERT predictions for BERTopic cluster keywords from 4chan and Endchan

Finally, the topic classification model provided by TweetNLP was applied. As shown in Table 4, *News & Social Concern* emerged as the most prevalent topic across both forums, which is expected given the political focus of the discussion boards analyzed in this report. In Figures 5 & 6 (Appendix), the relative topic distribution is visualized.

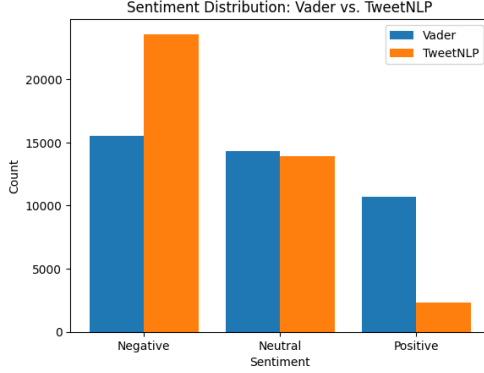
Topic	4chan Count	Endchan Count	4chan %	Endchan %
news_ & _social_ concern	322	114	60.30%	71.70%
diaries_ & _daily_ life	83	7	15.54%	4.40%
science_ & _technology	39	8	7.30%	5.03%
business_ & _entrepreneurs	25	7	4.68%	4.40%
sports	10	2	1.87%	1.26%

Table 4: Top 5 TweetNLP topics across 4chan and Endchan, with absolute counts and percentages.

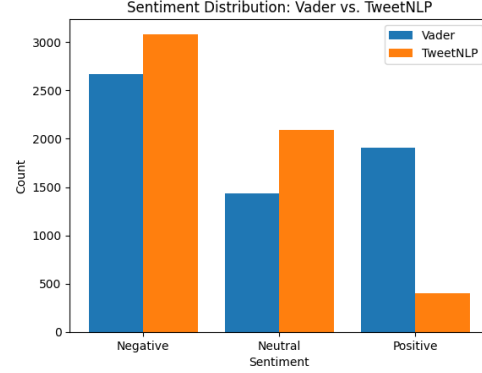
4.3 Sentiment Analysis

Sentiment Model Comparison The sentiment analysis was performed using VADER and TweetNLP to be able to compare the results of two different models on the same data. TweetNLP is a RoBERTa-based framework that provides a collection of models specifically trained on Twitter data for various natural language processing tasks [3]. For this project, we use their sentiment, emotion, irony, hate speech, and offensive language models. Furthermore, VADER was used as a baseline sentiment analysis model, consistent with the methodology used by Hine et al. [8].

Figures 2 and 7 present the sentiment distributions and agreement between VADER and TweetNLP for both 4chan and Endchan. We can see that TweetNLP consistently assigns a higher proportion of posts as negative and fewer as positive compared to VADER, indicating a stricter threshold for identifying positive sentiment. For example, in the 4chan dataset, TweetNLP labels over 23,000 posts as negative, while VADER assigns this label to just over 15,000. The confusion matrix further confirms this: 5,563 posts labeled positive by VADER are considered negative by TweetNLP. A similar pattern is observed in the Endchan dataset, where 737 posts marked as positive by VADER are labeled negative by TweetNLP. Despite some agreement in the neutral category (e.g., 8,578 posts on 4chan), the overall dissimilarity between the models highlights that model choice does significantly influence the sentiment classification.



(a) Sentiment distribution on 4chan



(b) Sentiment distribution on Endchan

Fig. 2

Interpreting Sentiment Through TweetNLP For further analysis, an interactive Streamlit dashboard is available at <https://dataforensics-3-sentimentforums.streamlit.app/>. The dashboard allows users to explore and compare sentiment analysis results across both forums. Users can select a dataset, perform crosstab analyses, and visualize the results as bar plots—either in absolute counts or percentages. Additionally, the interface enables filtering based on multiple TweetNLP model outputs (e.g., hate speech, emotion, and sentiment labels). For example, filtering for posts with no hate speech, the emotion “Anticipation,” and a positive sentiment will return the top five comments ranked by their associated confidence scores.

As observed from Figure 2, we can see that the negative and neutral sentiments dominate both forums. Still, to get more insight into the relationship between sentiment and the other classifications, a series of crosstab analyses were performed for both forums (Figures 8 & 9). Both figures show how sentiment correlates with the four classifications: emotion, hate speech, irony, and offensive language.

For both 4chan and Endchan, the negative sentiment is more associated with anger, disgust, and sadness, whereas positive sentiment is associated with joy and optimism, confirming a strong alignment between emotional tone and sentiment labels. Additionally, Endchan shows a slightly stronger dominance of anger in negative posts than 4chan.

In the hate speech analysis, it can be clearly seen that hate-labeled content is significantly more prevalent in negative sentiment posts. On 4chan, approximately 35% of negative posts contain hate speech, while on Endchan this proportion is slightly higher. Neutral and positive posts, in contrast, are mainly categorized as non-hate.

The irony distribution shows that ironic content is slightly more prevalent in the positive and negative sentiment rather than in the neutral sentiment. One possible reason for this might be that sarcasm or exaggeration may be more common in emotionally charged discussions. This interpretation is further supported by Figure 3, where irony frequently co-occurs with emotionally intense labels such as *anger*, *disgust*, *fear*, and *joy*.

Finally, in the offensive language analysis, it is shown that offensive content is mainly present in negative sentiment posts, whereas neutral and positive posts are mostly non-offensive. This distinction is more noticeable on Endchan, where less than 10% of neutral or positive posts are labeled offensive, compared to nearly 50% of negative ones.

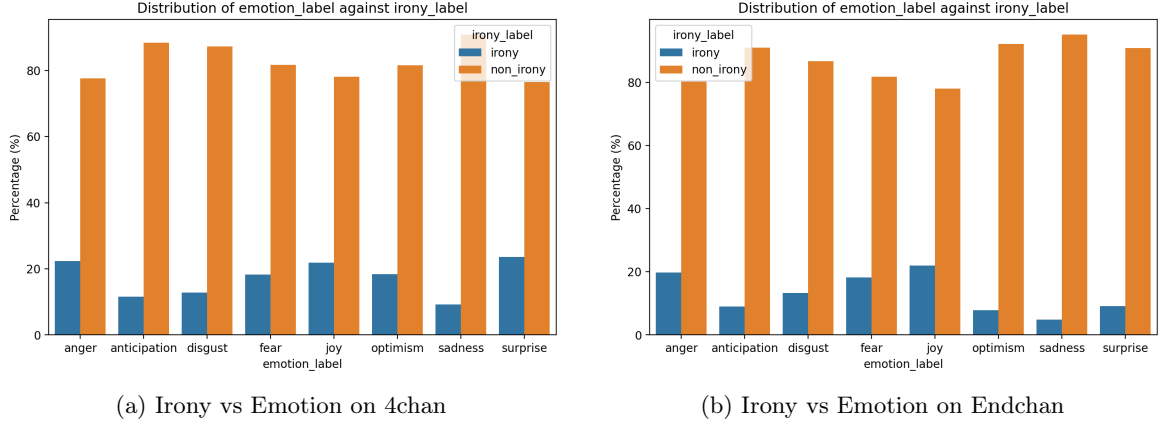


Fig. 3

4.4 Named Entity Recognition

To be able to analyze what was causing this positive sentiment, the NER was performed. This allowed the sentiment analysis per entity, for both of our forms. This sentiment analysis per entity once again showed interesting results. As can be seen in Figure 11 (Appendix), the entity 'the United States' has a very positive sentiment, especially when compared to all other entities.

Furthermore there are some entities that have an interestingly large negative sentiment on Endchan, such as 'Zionist', 'Israeli', and surprisingly 'Americans'. Surprising, because the outlier in extremely positive sentiment on Endchan is 'The United States', highly associated with 'Americans'. The extremely positive sentiment for 'The United States' is likely caused by American users discussing the history of their country in a positive light. However, foreigners will use the term 'Americans' to describe actions performed by Americans in a more negative light, as their aim is to criticize the country.

On 4chan, the negative sentiment is more strongly present than the positive sentiment, as can be seen in Figure 10 (in the Appendix). Some words have barely any positive sentiment, such as 'JFK', 'russians', and 'Christianity', and some words have an overall higher negative sentiment, such as 'Jesus', 'Russians', and 'jews'. When comparing these words to the words that have the lowest negative sentiment, such as 'Nazi', 'Germans', and 'Aryan' (a term associated with racism), it shows that this particular imageboard is most likely used to glorify a particular part of history that is usually viewed as a dark page in history books.

5 Discussion

The goal of this study was to explore the nature of political messages sent on anonymous forums by comparing two platforms: 4chan (surface web) and Endchan (dark web). Through topic modeling and sentiment analysis, we attempted to uncover differences in topics and sentiment between these forums.

5.1 Topic Modeling

At first, the topic modeling proved less informative than expected. Neither LDA nor BERTopic were able to generate coherent clusters with meaningful topic separation. The models combined with DarkBERT's unmasking

ability, although promising on dark web topics, struggled to generate meaningful topic labels. It often returned vague or circular terms (e.g., predicting “watch” as the topic of a cluster containing “watch,” “poll,” and “care”), underlining that dark web forums may need special models tailored to their contexts. So DarkBERT may not be trained enough for the context of dark web forums.

Another possible explanation for the underperformance of LDA and BERTopic could be that the input content was too homogeneous as all posts came from politically themed boards, which therefore limited the variation in topics. This explanation was then confirmed by TweetNLP’s topic modeling, which revealed the most common topic to be *News & Social concern*. This is most likely caused by the political nature of the imageboard. The topic modeling by TweetNLP does provide insightful results, showing that the model could potentially provide interesting conclusions when applied on a more neutral platform. Additionally, as noted by Hine et al. [8], these boards often use coded language, irony, and memes, which complicate conventional topic modeling.

Finally, these results are in line with observations made by Gehl [5] and Kajan [11], who noted that the anonymity and culture within those forums can result in context-dependent language, making automated content analysis less reliable and very model-dependent.

5.2 Sentiment Analysis

One of the most notable findings was the difference in sentiment classification between VADER and TweetNLP, as visualized in the distribution plots (Figure 2) and confusion matrices (Figure 7). TweetNLP consistently assigned more negative sentiment and significantly fewer positive classifications across both platforms. This result is likely due to the different modeling approaches: VADER is a rule-based model optimized for short, informal texts and tends to overestimate positivity, while TweetNLP (based on RoBERTa) is more nuanced and conservative in labeling content as positive.

This difference is especially evident in the confusion matrices, where a large number of posts labeled as positive by VADER were labeled as negative or neutral by TweetNLP. On 4chan, 5,563 posts marked positive by VADER were considered negative by TweetNLP, and on Endchan this occurred for 737 posts. This supports the hypothesis that VADER may be less sensitive to subtle negativity or sarcasm, which is common on politically incorrect boards. This aligns with De Gibert et al. [6], who noted the difficulty of classifying hate or offensive content using traditional tools. Our confusion matrices support this, showing that many posts labeled as positive by VADER were instead labeled negative by TweetNLP.

Furthermore, this difference in sentiments aligns with Hine et al. [8] and Zannettou et al. [17], who noted that the /pol/ boards on both forums attract extremist views. Our results, especially those from TweetNLP, seem to be in line with this observation: both 4chan and Endchan displayed strong negative sentiment patterns, especially toward political entities. In particular, our entity-level sentiment analysis revealed that terms like “Zionist,” “Israeli,” and even “Americans” were associated with strongly negative sentiment, especially on Endchan. This finding supports Topor’s [15] claim that platforms like Endchan serve as spaces for expressing right-wing antisemitic ideologies that are increasingly unwelcome in the mainstream media. These results demonstrate that model choice has a substantial impact on conclusions drawn from sentiment analysis, particularly in online environments where language may be coded, sarcastic, or intentionally ambiguous.

Finally, the most important part of our sentiment analysis was the development of an interactive Streamlit dashboard, which allowed for a dynamic approach to interpreting the data. Since users can dynamically filter and visualize posts based on sentiment, emotion, hate speech, irony, and offensive language classifications, they are able to conduct dynamic hypothesis testing for both forums. For example, users can isolate posts labeled as both

"angry" and "ironic" to explore comments with these undertones and examine these posts for linguistic patterns. The addition of model confidence score further allows users to select comments with greater certainty.

5.3 Limitations

A major limitation encountered during data collection was the imbalance in dataset sizes: the 4chan dataset was collected over a single day and consists of more than 40,000 posts, while the Endchan dataset spans several years but contains only around 6,000 posts. This difference may influence overall sentiment distribution and topic detection, especially given that newer posts may reflect different sociopolitical contexts than older ones. Future work could address this by sampling over comparable time windows or balancing the data.

Furthermore, there is room for improvement in future versions of the tool. Although both datasets contain timestamps, there is currently no option for temporal filtering, which could be added to help analyze changes in sentiment over time. Moreover, while users can select either forum to view results, comparing them side-by-side requires manually saving images. A valuable improvement would be enabling simultaneous loading or merging of datasets, so that visual comparisons between 4chan and Endchan can be made directly within the dashboard.

Finally, as noted by [3], the models used in this study are trained on Twitter data. However, there may still be subtle differences in linguistic patterns across various social media platforms. Therefore, future research could benefit from fine-tuning sentiment models specifically on imageboard or forum data, for example by using the labeled hate speech dataset from De Gibert et al. [6]. Nonetheless, our results demonstrate that using a more nuanced model leads to sentiment classifications that align more closely with theoretical expectations. In this regard, we build on the work of Hine et al. [8], who applied a basic NLTK model for sentiment analysis. We strongly recommend that future studies replicate their research using more advanced models, as we believe that re-analyzing their dataset with improved tools, such as interactive dashboards like ours, could result in even more insights.

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Appendix

A Data Overview

subpost_id	content	time	thread_id
91434	>>91417>	10/28/202	91417
91435	Netanyahu	10/28/202	91417
91448		10/29/202	91417
92023	ldk but it m	12/26/202	91417
92024	>>92023	12/26/202	91417
100017	>>97139>f	01/23/202	91417
100041	>>67551ht	01/23/202	91417
100052	>>82344O	01/23/202	91417
100053	>>14787H	01/23/202	91417
100078	>>4265	01/23/202	91417
100127	>>3469Yot	01/23/202	91417
100142	>>22649>f	01/23/202	91417
95331	We are: 10	12/08/202	95327
95334	Is Rand Pai	12/09/202	95327
95367	>>95331H	12/09/202	95327
95368	I'm being rc	12/09/202	95327
95374	>>95372>:	12/09/202	95327
95416	>>95367>f	12/10/202	95327
95426	>>95416>f	12/10/202	95327

(a) Example of the post dataset

thread_id	thread_text	time	
91417	What's wr	10/27/2023 (Fri)	15:50
95327	1) disable	12/08/2024 (Sun)	23:36
93932	They like to	06/15/2024 (Sat)	11:36
90261	The Natasl	04/14/2023 (Fri)	02:05
94806	Hello from	09/28/2024 (Sat)	02:57
92142	General, c	01/06/2024 (Sat)	13:04
82014	Germania	09/01/2020 (Tue)	17:04
88984	thread of s	10/11/2022 (Tue)	19:45
86576	Calls for p	02/27/2022 (Sun)	00:20
90586	As Americ:	06/05/2023 (Mon)	22:07
90730	FOOTAGE:	07/09/2023 (Sun)	15:45
102561	This is the	03/03/2025 (Mon)	00:41
94245	it's hard be	07/28/2024 (Sun)	06:54
76282	>White sup	10/22/2019 (Tue)	18:28
91167	The Punish	09/29/2023 (Fri)	23:23
95171	I found a di	11/27/2024 (Wed)	22:54
94740	The Art Of	09/20/2024 (Fri)	18:15
76630	Zelda is Fri	11/08/2019 (Fri)	08:25
93081	Interesting	04/05/2024 (Fri)	01:48
77267	ITT: We del	11/27/2019 (Wed)	15:20

(b) Example of the threads dataset

Fig. 4

B Topic Modeling

B.1 NLTK + LDA Topic Modeling

#	Topic 0 words w.		Topic 1 words w.		Topic 2 words w.		Topic 3 words w.		Topic 4 words w.	
0	https	128.2	us	42.1	make	70.4	even	39.5	white	78.5
1	trump	120.0	go	35.0	like	45.9	www	36.0	would	61.3
2	com	101.2	europe	30.2	get	23.1	uk	33.2	people	36.1
3	www	55.4	's	26.4	year	18.4	new	26.2	world	29.7
4	year	14.0	like	18.5	would	7.3	's	19.8	literally	27.2
5	every	10.4	uk	7.3	people	6.7	full	18.5	government	14.7
6	uk	9.1	world	5.7	white	0.9	go	15.4	even	9.9
7	make	5.3	www	0.2	even	0.2	people	12.9	every	9.5
8	full	4.9	year	0.2	go	0.2	make	11.9	get	8.3
9	government	4.8	would	0.2	government	0.2	government	8.1	full	4.2
10	's	2.9	even	0.2	every	0.2	every	7.6	us	2.3

Table 5: NLTK: Top words and their weights across 5 topics for 4chan

#	Topic 0 words w.		Topic 1 words w.		Topic 2 words w.		Topic 3 words w.		Topic 4 words w.	
0	com	97.2	jews	125.1	's	76.3	world	65.0	people	61.2
1	https	91.9	jewish	98.7	many	17.5	's	28.1	would	41.8
2	one	7.5	world	15.0	like	17.4	time	22.7	like	36.3
3	like	6.7	would	8.6	even	4.8	people	16.3	one	26.3
4	many	5.8	even	7.7	one	4.3	one	11.9	even	22.8
5	world	1.6	time	7.1	jewish	3.7	many	3.3	time	15.8
6	even	0.4	many	4.7	people	2.8	https	0.4	many	5.7
7	jews	0.3	one	2.9	https	2.2	jewish	0.2	jews	0.2
8	time	0.2	like	1.4	com	2.2	like	0.2	https	0.2
9	jewish	0.2	people	0.5	would	0.2	com	0.2	jewish	0.2
10	people	0.2	https	0.2	time	0.2	would	0.2	world	0.2

Table 6: NLTK: Top words and their weights across 5 topics for Endchan

B.2 BERTopic

Topic	Count	Name	Representation
0	31	-1_make_woods_japanese_live	[make, woods, japanese, live, samurai, escape, ...]
1	37	0_white_search_google_anti	[white, search, google, anti, nobody, men, fam...]
2	36	1_recall_european_would_tesla	[recall, european, would, tesla, glue, musk, e...]
3	34	2_notice_courts_something_libs	[notice, courts, something, libs, slated, acce...]
4	31	3_india_prison_china_lockdown	[india, prison, china, lockdown, manufacturing...]
5	29	4_faces_sold_want_even	[faces, sold, want, even, child, friends, make...]
6	27	5_step_great_every_literal	[step, great, every, literal, america, endless...]
7	26	6_care_baltics_fuck_russians	[care, baltics, fuck, russians, whitebaltics,...]
8	25	7_ukraine_bbc_uk_catbox	[ukraine, bbc, uk, catbox, moe, news, co, arti...]

Topic	Count	Name	Representation
9	22	8_poliivere_polls_policy_real	[poliivere, polls, policy, real, never, pol, c...]
10	21	9_convicted_shitailans_tomatokikes	[kosher, convicted, shitailans, tomatokikes, semitic,...]

Table 7: BERTopic-generated topics from 4chan data (excluding representative documents)

Topic	Count	Name	Representation
0	65	-1_com_www_https_israel	[com, www, https, israel, war, world, russian,...]
1	42	0_people_want_media_like	[people, want, media, like, first, thread, ame...]
2	28	1_jews_jewish_hitler_world	[jews, jewish, hitler, world, jew, make, ritua...]

Table 8: BERTopic-generated topics from Endchan data (excluding representative documents)

B.3 TweetNLP Topic Modeling

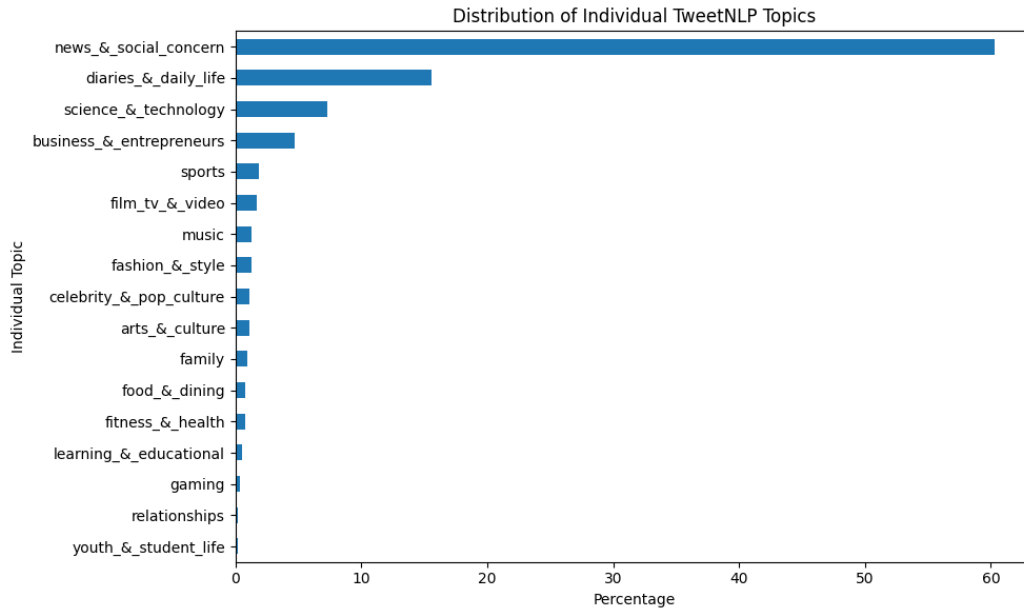


Fig. 5: TweetNLP topic distribution on 4chan

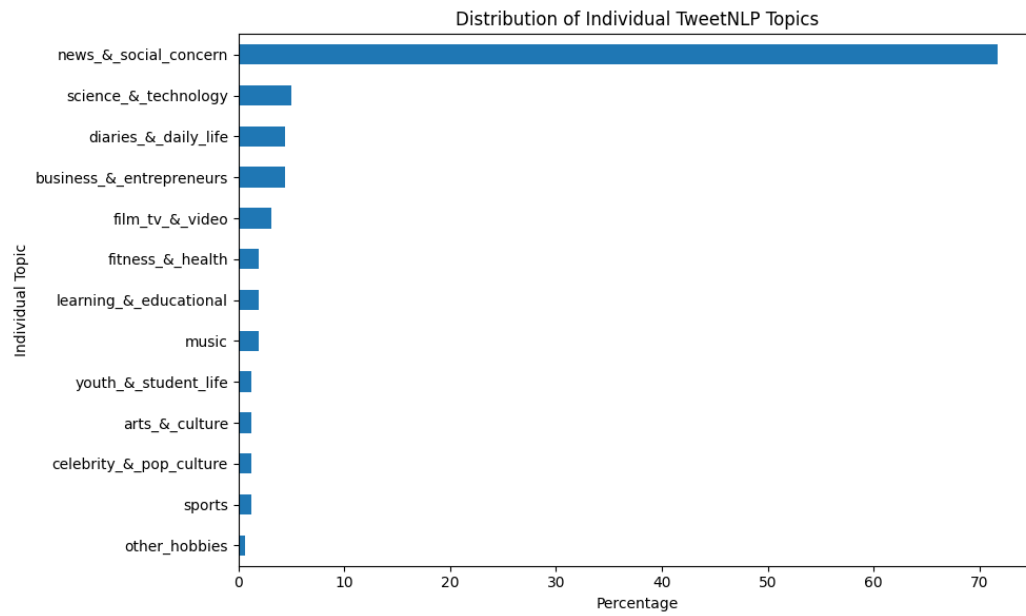
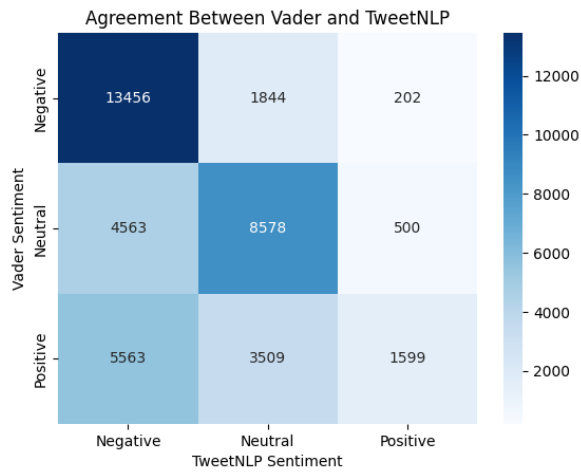


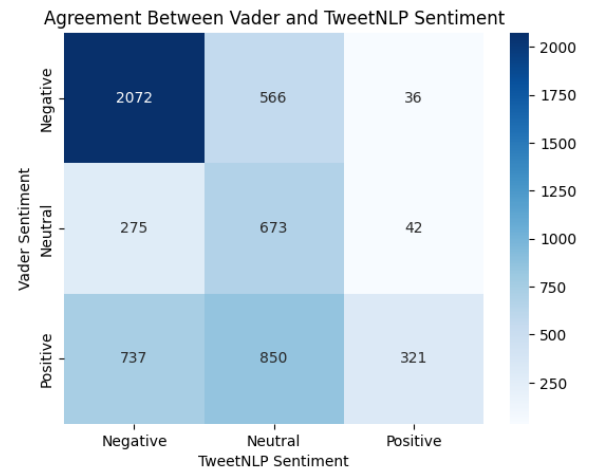
Fig. 6: TweetNLP topic distribution on Endchan

C Sentiment Analysis

C.1 Vader VS TweetNLP



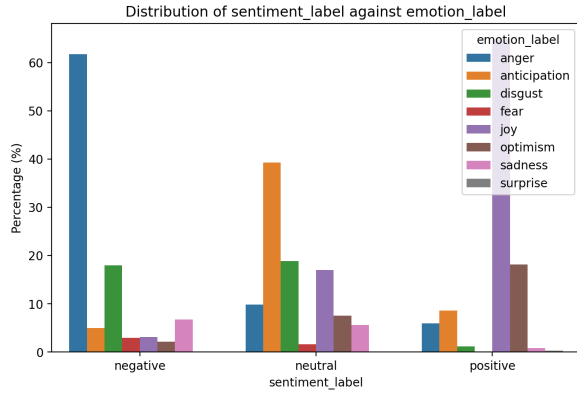
(a) Confusion matrix for sentiment on 4chan



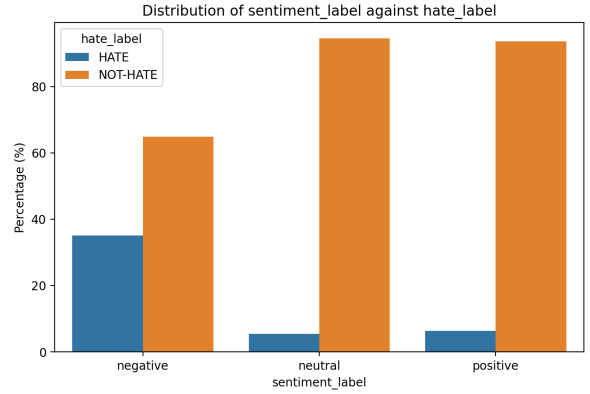
(b) Confusion matrix for sentiment on Endchan

Fig. 7

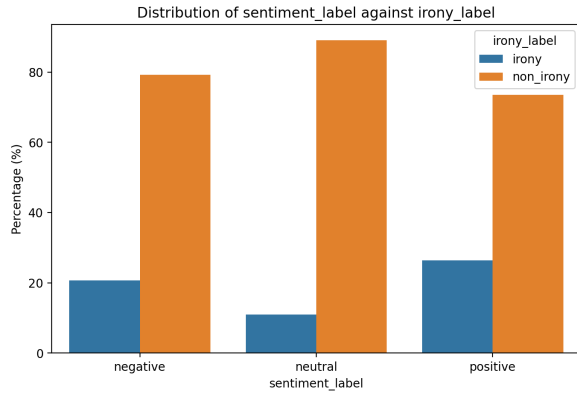
C.2 TweetNLP Crosstab Analysis



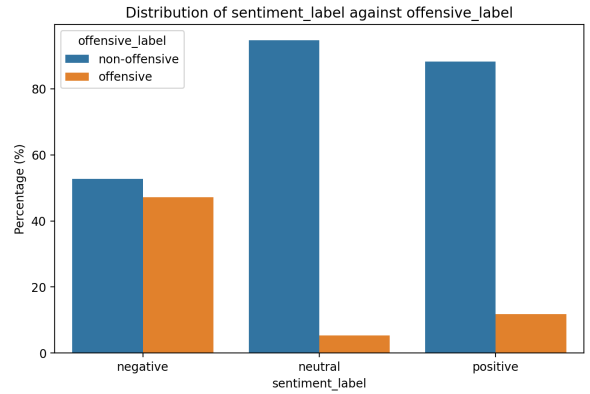
(a) 4chan: Sentiment against Emotion



(b) 4chan: Sentiment against Hate

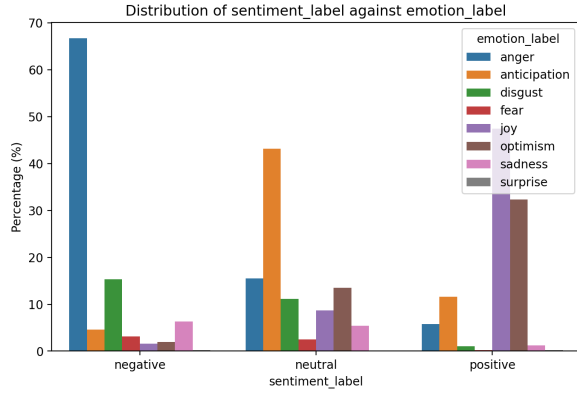


(c) 4chan: Sentiment against Irony

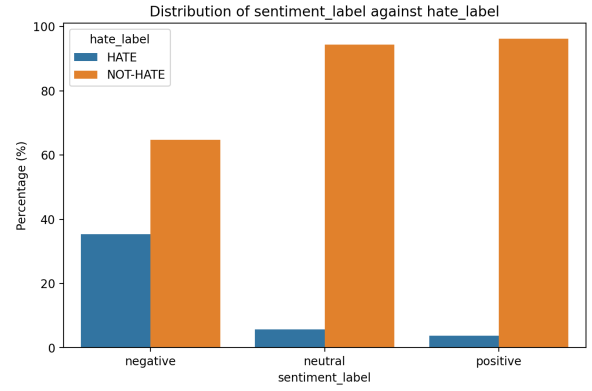


(d) 4chan: Sentiment against Offensive

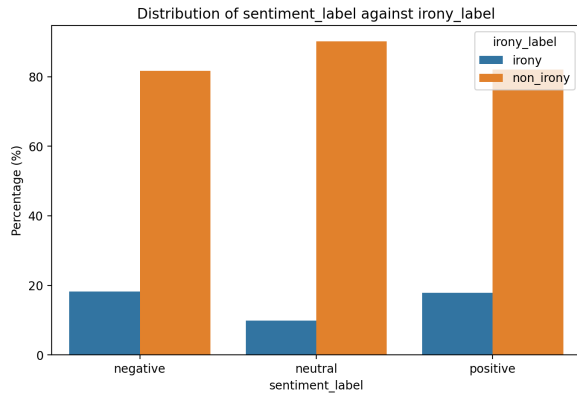
Fig. 8: Overview of Crosstab sentiment analysis for 4chan



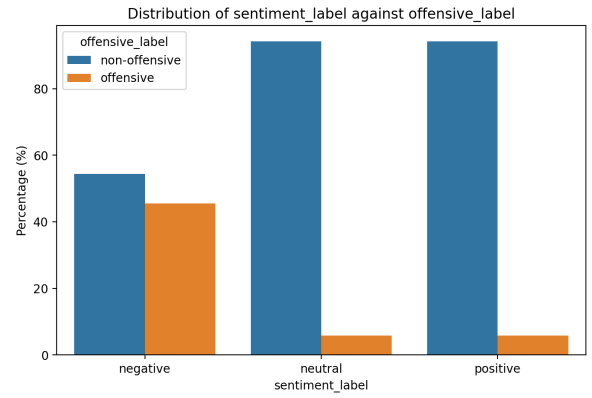
(a) Endchan: Sentiment against Emotion



(b) Endchan: Sentiment against Hate



(c) Endchan: Sentiment against Irony



(d) Endchan: Sentiment against Offensive

Fig. 9: Overview of Crosstab sentiment analysis for Endchan

D Named Entity Recognition



Fig. 10: Overview of sentiment of top 50 4chan entities

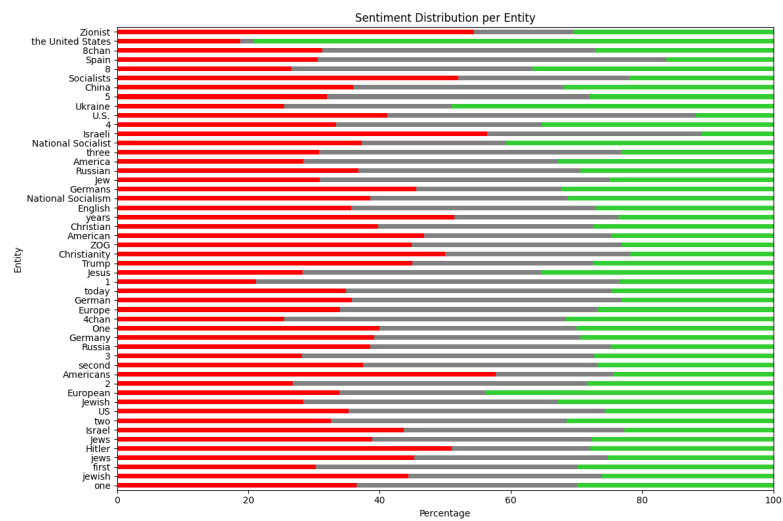


Fig. 11: Overview of sentiment of top 50 Endchan entities